Project Title: Graduation Dropout Prediction

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Problem Description:
We aim to predict student dropout rates using supervised machine learning. Our primary task is a regression problem predicting PctDropouts. We will also explore a classification task that labels subgroups as high- or low-risk based on a dropout threshold. Student dropout is a major educational and socioeconomic issue linked to lower lifetime earnings and employment instability. Identifying high-risk demographic and institutional subgroups can support targeted interventions and data-driven educational policy. Machine learning enables us to capture nonlinear patterns and interactions across race, gender, grade level, and district characteristics.

Dataset: Delaware Open Data Portal – Student Dropout Dataset
https://data.delaware.gov/Education/Student-Dropout/7u9f-65y3/about_data

- ~91,900 rows, 15 columns
- Each row represents dropout statistics for a subgroup defined by race, gender, grade, special demographic status, geography, and organization level
- Key features: categorical (Race, Gender, Grade, SpecialDemo, Geography, District), numerical (Dropouts, Students, PctDropouts), and School Year
- Redacted rows will be removed

The dataset is sufficiently large to support meaningful modeling and comparison across methods.

Approach and Methodology:
Feature Normalization & Preprocessing:

- Remove redacted rows and handle missing values
- One-hot encode categorical variables
- Standardize numerical features using scaling
- Engineer additional features (e.g., dropout rate per 100 students, year trends if applicable)

Models: Linear Regression, Logistic Regression, Random Forest, Gradient Boosting, Neural Network (MLP)

Tools: Python, Pandas, NumPy, Scikit-learn, TensorFlow/PyTorch, Matplotlib.

Evaluation:

- Regression: MSE, RMSE, R^2
- Classification: Accuracy, Precision, Recall, F1, ROC-AUC
- Cross-validation for robustness

Outcome: We aim to build an accurate and interpretable model that identifies high-risk subgroups and key predictive factors. We expect to compare model performance, analyze feature importance, and provide insights that could inform educational policy and intervention strategies.

Plan/Proposed Split:
Jessica: data cleaning, feature engineering, linear/logistic regression, evaluation.
Aarushi: tree-based models, neural network, hyperparameter tuning, and visualization.
Both: experimentation, comparison, and final report.